

Chapter 30: The Monetary System

If banks can create money, why can't they make us all rich?
Consider what gives money value and how the banking
system actually increases the money supply.



Learning Objectives

30.1 Explain the functions and forms of money.

30.2 Describe the structure and roles of the Federal Reserve System.

30.3 Analyze how banks create money in a fractional reserve system.

30.4 Evaluate how the Federal Reserve controls the money supply.

30.5 Discuss the limits of monetary control and implications for the economy.

Key Points

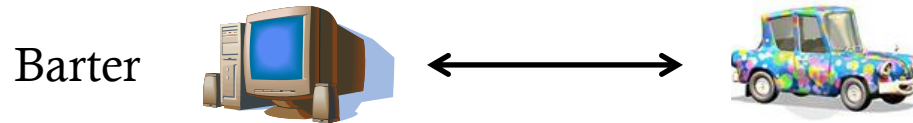
- Money consists of assets people use to buy goods and services. It performs three essential functions:
- **Medium of exchange** for transactions,
- **Unit of account** for measuring values, and
- **Store of value** for transferring purchasing power over time.
- Money can take the form of **commodity money** (with intrinsic value, like gold) or **fiat money** (without intrinsic value, like the U.S. dollar). In the United States, money includes currency and bank deposits such as checking accounts.
- The **Federal Reserve (the Fed)** serves as the nation's central bank, overseeing the monetary system and setting monetary policy through the **Federal Open Market Committee (FOMC)**.

Key Points

- Banks influence the money supply by holding only a fraction of deposits as reserves and lending the rest, which creates new money. Because of this, the Fed's control over the money supply is not perfect.
- Banks operate with **capital** supplied by owners and use **leverage** to expand assets, but this also increases risk. To protect depositors, regulations require minimum capital levels.
- The Fed manages the money supply using tools such as **open-market operations**, **the discount rate**, **reserve requirements**, and **interest on reserves**. By adjusting these, it can expand or contract the supply of money. Since 2008, interest on reserves has become a key tool.
- Today, the Fed conducts monetary policy mainly by targeting the **federal funds rate**—the interest rate at which banks lend to one another overnight—adjusting the money supply to achieve its target.

The Purpose and Function of Money

1. **Medium of exchange** – used to buy goods and services



Reduces transaction costs

Money allows for specialization

2. **Store of value** – can be held for future purchases

Liquidity

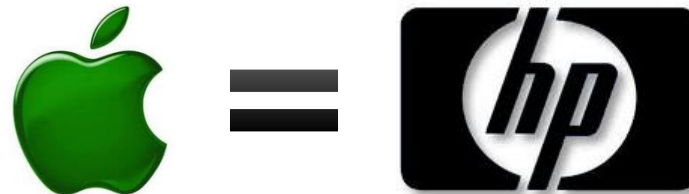
Least exclusive function



Ahh!!!
Economics

3. **Unit of account** – allows products to be quoted in terms of others

Relative Prices



Components of Money

M1



1) Currency

- a. **Token Money** - intrinsic value is less than actual value
- b. Federal Reserve Notes

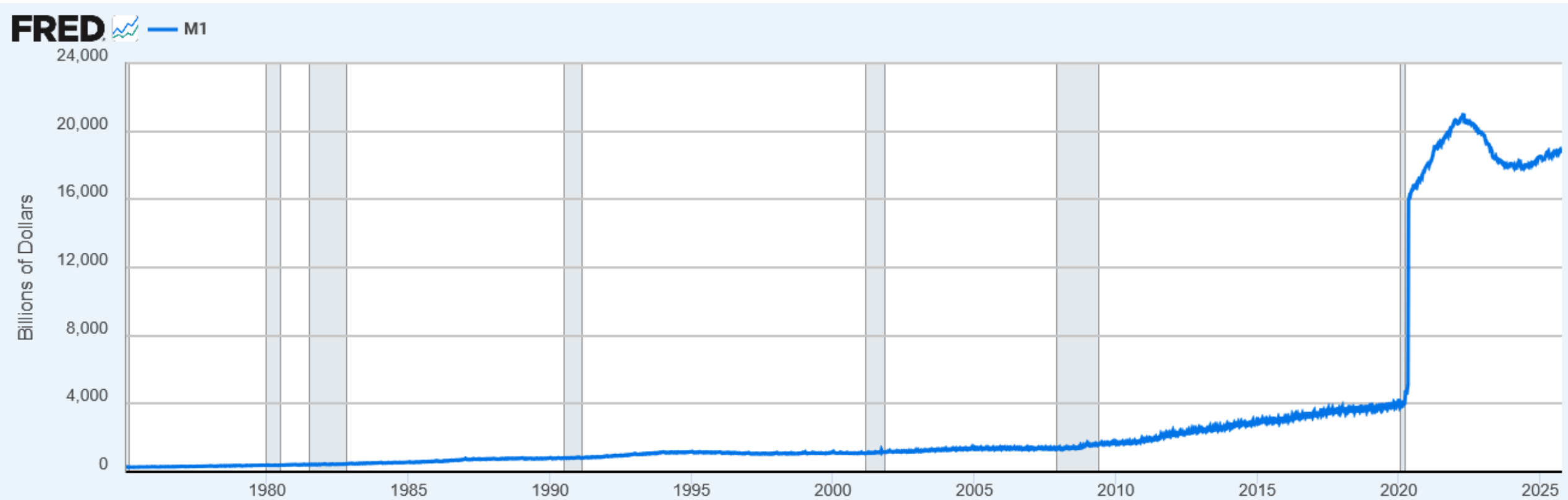
2) Checkable Deposits

3) Traveler's Checks

- a. Are not as important as it used to be

If it's held by the federal government, Federal Reserve, or financial institution it does not count

M1



Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

How much does it cost to mint our coins and dollars?

- The Coinage Act of 1792 authorized the US mint to produce coins (copper, silver, and gold) for circulation.
- Today, metals such as zinc and copper determine the bulk of the cost of coins
- For 9 straight fiscal years the penny and nickel have been above their face value.

	<u>Penny</u>	<u>Nickel</u>	<u>Dime</u>	<u>Quarter</u>	<u>Dollar</u>
Cost of Goods Sold	\$0.0178	\$0.0659	\$0.0323	\$0.0778	
Selling, General & Administrative	\$0.0025	\$0.0085	\$0.0045	\$0.0099	
Distribution to FRB	\$0.0003	\$0.0009	\$0.0005	\$0.0011	
Total Unit Cost	\$0.0206	\$0.0753	\$0.0373	\$0.0887	\$0.055

Source: usmint.gov, federalreserve.gov

Components of Money

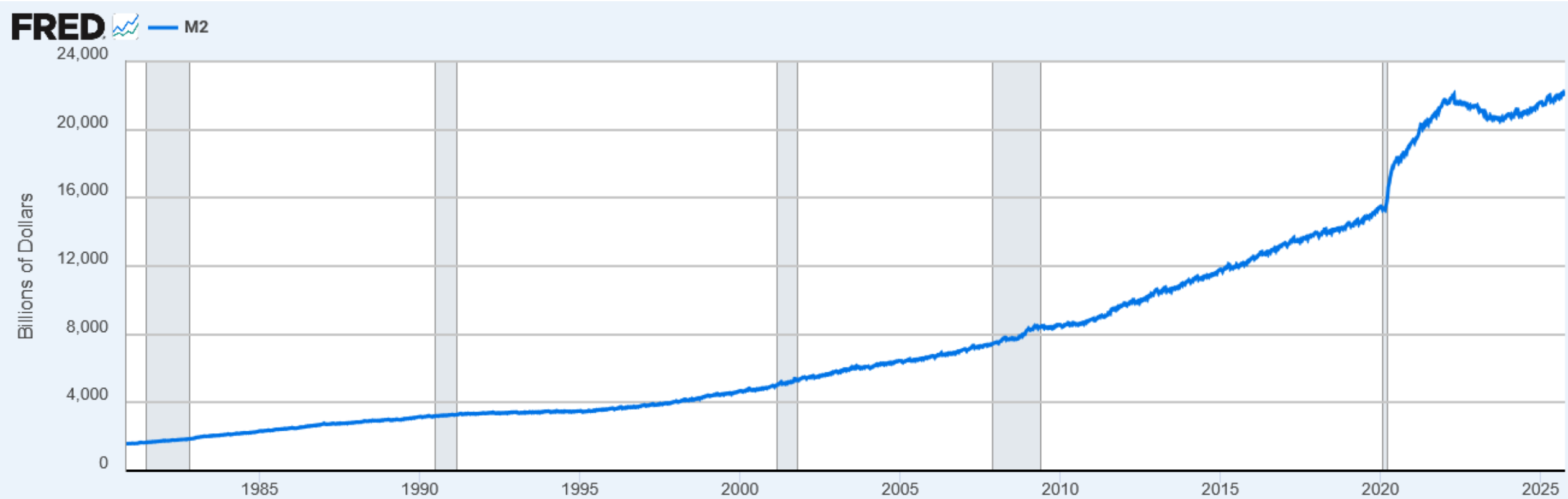
M2

M1 + “Near Monies”

Near Monies

- 1) Savings deposits and money market deposit accounts (MMDAs)
 - a. Interest earning
 - b. MMDA – limited check writing
- 2) Time Deposits < \$100,000
 - a. Interest earning
 - b. Penalties
- 3) Money Market Mutual Funds
 - a. Interest earning
 - b. Retail MMMFs

M2



Source: Board of Governors of the Federal Reserve System (US) via FRED®
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Activity: What is M1 and M2?

Using the table below, answer these questions:

1. What is the size of the M1 money supply?
2. What is the size of the M2 money supply?
3. What is the value of the M2 money supply not counted in the M1 money supply?

Item	Billions of Dollars
Checkable Deposits	\$700
Small Time Deposits	\$900
Currency	\$650
MMMF held by businesses	\$1,750
Savings Deposits including MMDAs	\$3,000
MMMF held by individuals	\$1,000

Activity: What is M1 and M2?

1. Checkable Deposits + Currency:

- $\$700 + 650 = \1350

2. M1 + Small Time Deposits + Savings Deposits + MMMF held by individuals:

- $(\$700 + 650) + (\$1,000 + 3,000 + 900) = \$6,250$

3. M2 – M1 (near monies):

- $\$6,250 - (\$700 + 650) = \$4,900$

Money Supply

What backs the money supply?

- 1) Gold ?
 - a. Gold Standard
 - b. Bretton Woods System (1971-1973)
 - c. Wizard of Oz
- 2) Faith in the US government
 - a. Ability to keep value stable
- 3) Money is scarce
 - a. Supply and demand determine its scarcity

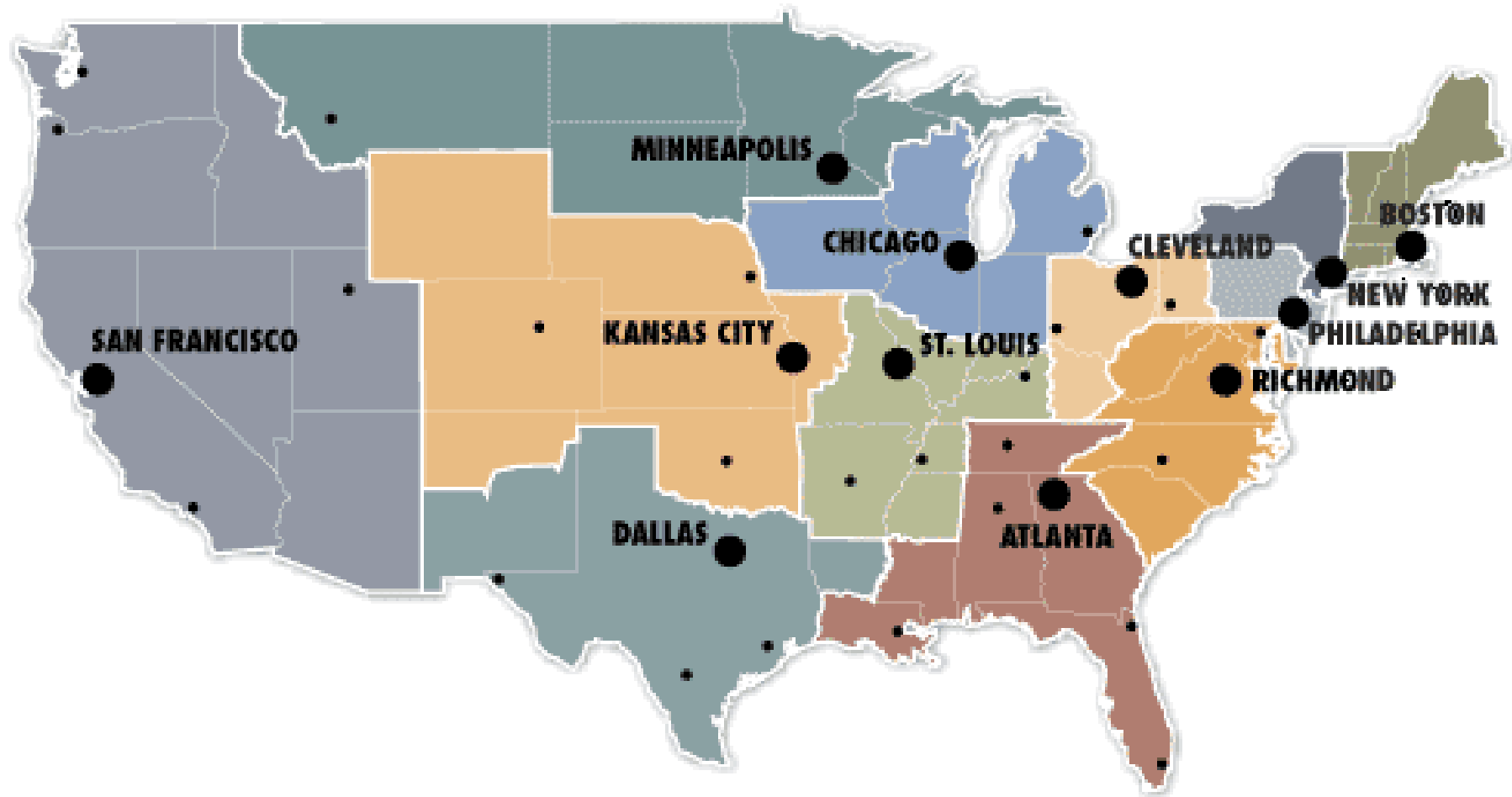
Money Supply

- Money's Value
 - Intrinsic valuation
 - **Fiat Money** – no intrinsic value
 - **Commodity Money** – takes the form of a commodity such as gold.
- Purchasing Power ($V\$ = 1/P$)
 - Price level increase causes money to lose value
 - Price level decrease causes money to gain value
- **Hyperinflation** – rapid rise in the price level
 - Ex: Germany post WWI, Hungary 1945, Venezuela 2010s

The Federal Reserve

- The Federal Reserve Act (1913)
- The “Fed” opened for business (November 1914)
- **Operations**
 1. Board of Governors
 1. 7 members each serving 14-year terms
 2. Each member is elected by the President and confirmed by the Senate, and they serve on a 2-year rotating basis
 3. Who is the head of the Board of Governors?
 2. Federal Open Market Committee
 - 7 Governors + 5 regional presidents
 - Buying and selling bonds is a primary function
 - Monetary Policy is conducted via the FOMC
 - New York Fed is where buying/selling bonds takes place

The Federal Reserve



Functions of the Fed

1. Issuing “Federal Reserve Notes”
2. Sets reserve requirements and holds the reserves
3. Lending Money “Lender of Last Resort”
 - **Discount rate** – interest rate the Fed charges banks for loans
4. Check collection and clearing service for banks
5. Federal Reserve System acts as the fiscal agent for the Federal government
6. The Federal Reserve System supervises member banks
7. Conducts monetary policy and controls the money supply

Other important notes:

- Federal Reserve Independence
- Private Bank owned by its member banks
- Free from political pressure

Bank Creation

- Creation begins with customers/owners giving money to someone to create the bank
- The result is shown in a balance sheet or T-account
- A customer gives deposits \$1000 in my bank

Assets		Liabilities and net worth	
Cash	\$1000	Stock	\$1000

- **Vault Cash** – cash held by the bank
- **Assets** – anything of monetary value owned by a firm or individual
- **Liabilities** – a debt(s) with monetary value; owed by a firm or individual
- **Net Worth** – total assets less total liabilities of a firm or individual (claims)

100% Reserve System

Assets		Liabilities and net worth	
Reserves	\$1000	Deposits	\$1000

- **Demand Deposits** – balances in bank accounts that depositors can access by writing a check
- **Reserves** – deposits the bank has received but has not loaned out
- Is the balance sheet in balance?
- Has the money supply increased?

Money Creation

- Troy Bank & Trust lends out all checkable deposits
- The “Fed” and Required Reserves
- Reserves go to Federal Reserve or is kept in vaults “vault cash”
- **Reserves** – Bank Deposits at the Fed + Vault Cash
- **Required Reserves** = $r * \text{Checkable Deposits}$

What's the point of reserves?

- Limits money creation
- **Reserve Types:**
 - **Actual Reserves** – what the bank has on hand
 - **Required Reserves** – the amount that a bank must keep at a Fed bank or what it holds at its bank as set by the Fed
 - **Excess Reserves** – (actual reserves – required reserves)
 - The role of excess reserves is that it allows banks to loan out money.
 - This is how the fractional reserve system works
- **Fractional Reserve Banking** - a reserve requirement that is less than 100% of the checkable-deposits liabilities of a commercial bank.

Fractional Reserve System at Work

- Suppose now our bank loans out the excess reserves it has on hand. First, assume that the reserve ratio $(r) = 10\%$
- How much can the bank lend?
- Assume the bank lends all its excess reserves, does the money supply increase?

Assets		Liabilities and net worth	
Reserves	\$1,000	Checkable Deposits	\$1,000

Fractional Reserve System at Work

- Excess Reserves = $\$1000 - (0.10 \times 1,000) = \900
- Yes, by \$900

Assets		Liabilities and net worth	
Reserves	\$100	Checkable Deposits	\$1,000
Loans	\$900		

Activity: Bank Lending

1. Suppose the reserve ratio is 10%, and the bank deposits all its cash with the Fed; how much can the bank lend?
2. If the reserve ratio is 25%; how much are excess reserves? How much can the bank lend?

Assets		Liabilities and net worth	
Cash	\$0	Checkable Deposits	\$60,000
Reserves	\$_____	Stock	\$90,000
Loans	\$_____		
Property	\$90,000		

Activity: Bank Lending

1. $\$54,000 = \$60,000 - (0.10 \times \$60,000)$ – see the table below
2. $\$45,000 = \$60,000 - (0.25 \times \$60,000)$
 1. Required reserves $= 0.25 \times \$60,000 = \$15,000$

Assets		Liabilities and net worth	
Cash	\$0	Checkable Deposits	\$60,000
Reserves	\$6,000	Stock	\$90,000
Loans	\$54,000		
Property	\$90,000		

Activity: Increasing Reserve Requirements

1. If the bank faces a reserve requirement of 10 percent, how much are the bank's excess reserves?
2. If the bank faces a reserve requirement of 20 percent, how much are the bank's excess reserves?

The First Bank of Troy

Assets		Liabilities	
Reserves	\$50,000	Deposits	\$300,000
Loans	\$200,000		

Activity: Increasing Reserve Requirements

1. If the bank faces a reserve requirement of 10 percent, how much are the bank's excess reserves?
 1. $\$300,000 \times 0.10 = \$30,000$
 2. $\$300,000 - \$30,000 = \$270,000$
 3. $\$270,000 - \$200,000 = \$70,000$
2. If the bank faces a reserve requirement of 20 percent, how much are the bank's excess reserves?
 1. $\$300,000 \times 0.20 = \$60,000$
 2. $\$300,000 - \$60,000 = \$240,000$
 3. $\$240,000 - \$200,000 = \$40,000$

Securities and Fractional Reserve Systems

Assets		Liabilities and net worth	
Reserves	\$60,000	Checkable Deposits	\$50,000
Property	\$90,000	Stock	\$100,000

- The bank buys US Treasury Bonds

Assets		Liabilities and net worth	
Reserves	\$60,000	Checkable Deposits	\$75,000
Property	\$25,000	Stock	\$100,000
Property	\$90,000		

- Assets increase by \$25,000 and checkable deposits do as well as it credits the account of the seller.

Securities and Fractional Reserve Systems

- Buying Securities
 - Banks and the Fed
 - Money Supply Increases
- Selling Securities
 - Banks and the Fed
 - Money Supply Decreases



The Banking System

One bank: \$\$\$

All banks : \$

Multiple deposit expansion

Assumptions

- 1) All banks have a set required reserve ratio
- 2) All banks are loaned up. They have no excess reserves
- 3) When a bank gains excess reserves, it loans it all out. Repeat...

The Simple Money Multiplier

Bank 1

Reserve ratio (r) = 10%

Assets		Liabilities and net worth	
Reserves	\$10	Checkable Deposits	\$100
Loans	\$90		

Bank 2 receives the loan and adjusts their balance sheet

Assets		Liabilities and net worth	
Reserves	\$9	Checkable Deposits	\$90
Loans	\$81		

Bank 3 receives the next loan and adjusts its balance sheet

Assets		Liabilities and net worth	
Reserves	\$8.10	Checkable Deposits	\$81
Loans	\$72.90		

Multiple Deposit Expansion

- A new deposit of \$100 and a reserve requirement of 10% generates \$900.

Bank	New Deposits	Required Reserves	Deposits Created (Loans)
A	100	10.00	90.00
B	90.00	9.00	81.00
C	81.00	8.10	72.90
D	72.90	7.29	65.61
E	65.61	6.56	59.049
⋮	⋮	⋮	⋮
Total	1,000	100	900

- Will a bank always loan out all its excess reserves?

Multiple Deposit Expansion

- From the previous table, more money created with each successive loan
- Why? Reserves lost = Reserves gained
- m = money multiplier
- How does the multiplier expand the money supply?
- Multiple deposit expansion (**D**)
- **$D = ER * m$**

$$m = \frac{1}{\text{required reserve ratio}} = m = \frac{1}{r}$$

Multiple Deposit Expansion Examples

Example 1

- $r = 10\%$
- $m = 1 / r \rightarrow m = 1 / 0.10 \rightarrow m = 10$
- Excess Reserves (ER) = **\$1,000**
- $D = \$1,000 \times 10 = \textbf{\$10,000}$

Example 2

- Checkable deposits = \$400,000; reserves = \$150,000; $r = 25\%$
- How much can the bank system loan out?
- Single bank : **\$300,000** = $\$400,000 - 0.25 \times \$400,000$
- All banks : **\$1,200,000** = $\$300,000 \times (1/0.25)$

A More Realistic Balance Sheet

- Assets: Besides reserves and loans, banks also hold securities.
- Liabilities: Besides deposits, banks also obtain funds from issuing debt and equity.
- **Bank capital**: the resources a bank obtains by issuing equity to its owners
 - Also: bank assets minus bank liabilities
- **Leverage**: the use of borrowed funds to supplement existing funds for investment purposes

A More Realistic Balance Sheet

MORE REALISTIC NATIONAL BANK

Assets		Liabilities	
Reserves	\$ 200	Deposits	\$ 800
Loans	\$ 700	Debt	\$ 150
Securities	\$ 100	Capital	\$ 50

- **Leverage ratio:** the ratio of assets to bank capital
- In this example, the leverage ratio = $\$1000/\$50 = 20$
- Interpretation: for every \$20 in assets,
\$ 1 is from the bank's owners,
\$19 is financed with borrowed money.

Leverage Amplifies Profits and Losses

- In our example, suppose bank assets appreciate by 5%, from \$1000 to \$1050. This increases bank capital from \$50 to \$100, doubling owners' equity.
- Instead, if bank assets decrease by 5%, bank capital falls from \$50 to \$0.
- If bank assets decrease more than 5%, bank capital is negative and bank is insolvent.
- **Capital requirement:** a government regulation that specifies a minimum amount of capital, intended to ensure banks will be able to pay off depositors and debts.

The Fed's Tools of Monetary Control

- Earlier, we learned $\text{money supply} = \text{money multiplier} \times \text{bank reserves}$
- The Fed can change the money supply by changing bank reserves or changing the money multiplier.

How the Fed Influences Reserves

Open-Market Operations (OMOs) – the purchase and sale of U.S. government bonds by the Fed.

- If the Fed buys a government bond from a bank, it pays by depositing new reserves in that bank's reserve account.

With more reserves, the bank can make more loans, increasing the money supply.

- To decrease bank reserves and the money supply, the Fed sells government bonds.

How the Fed Influences Reserves

- The Fed makes loans to banks, increasing their reserves.
 - Traditional method: adjusting the **discount rate**—the interest rate on loans the Fed makes to banks—to influence the amount of reserves banks borrow
- The more banks borrow, the more reserves they have for funding new loans and increasing the money supply.
- [Current list of policy tools](#)
 - **Term Asset-Backed Securities Loan Facility** (implemented March 23, 2020)
 - will enable the issuance of asset-backed securities (ABS) backed by student loans, auto loans, credit card loans, loans guaranteed by the Small Business Administration (SBA), and certain other assets.
 - Under the TALF, the Federal Reserve will lend on a non-recourse basis to holders of certain AAA-rated ABS backed by newly and recently originated consumer and small business loans. The Federal Reserve will lend an amount equal to the market value of the ABS less a haircut and will be secured at all times by the ABS.

How the Fed Influences the Reserve Ratio

- Recall: $\text{reserve ratio} = \text{reserves} / \text{deposits}$, which inversely affects the money multiplier.
- The Fed sets **reserve requirements**: regulations on the minimum amount of reserves banks must hold against deposits.
Reducing reserve requirements would lower the reserve ratio and increase the money multiplier.
- Since 10/2008, the Fed has paid interest on reserves banks keep in accounts at the Fed.
 - Raising this interest rate would increase the reserve ratio and lower the money multiplier.

Problems Controlling the Money Supply

- If households hold more of their money as currency, banks have fewer reserves, make fewer loans, and money supply falls.
- If banks hold more reserves than required, they make fewer loans, and money supply falls.
- Yet, Fed can compensate for household and bank behavior to retain fairly precise control over the money supply.

Case: Coin Shortage

Why do U.S. coins seem to be in short supply?

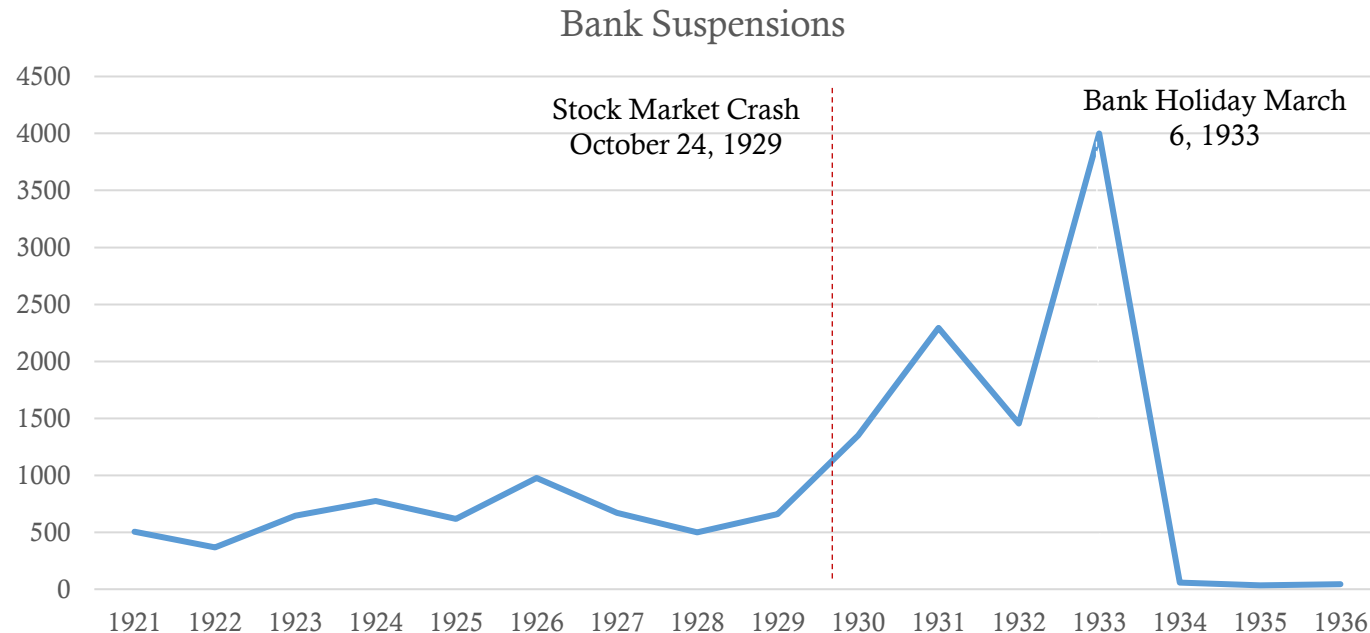
- Business and bank closures associated with the COVID-19 pandemic have significantly disrupted the supply chain and normal circulation patterns for U.S. coins. While there is an adequate overall amount of coins in the economy, the slowed pace of circulation has reduced available inventories in some areas of the country.
- The Federal Reserve is working with the U.S. Mint and others in the industry on solutions. As a first step, a temporary cap was imposed on the orders depository institutions place for coins with the Federal Reserve to ensure that the current supply is fairly distributed. In addition, a U.S. Coin Task Force was formed to identify, implement, and promote actions to address disruptions to coin circulation.
- Since mid-June, the U.S. Mint has been operating at full production capacity, minting almost 1.6 billion coins in June and is on track to mint 1.65 billion coins per month for the remainder of the year.
- As the economy recovers and businesses reopen, more coins will flow back into retail and banking channels and eventually into the Federal Reserve, which should allow for the rebuilding of coin inventories.
- [Reference: FederalReserve.gov](https://www.federalreserve.gov)

Bank Runs and the Money Supply

- A **run on banks**:
When people suspect their banks are in trouble, they may “run” to the bank to withdraw their funds, holding more currency and less deposits.
- Under fractional-reserve banking, banks don’t have enough reserves to pay off ALL depositors, hence banks may have to close.
- Also, banks may make fewer loans and hold more reserves to satisfy depositors.
- These events increase *reserve ratio*, reverse the process of money creation, cause money supply to fall.

Application: Banking Panics of 1930-31

- In the fall of 1930, the economy appeared poised for recovery. The previous three contractions, in 1920, 1923, and 1926, had lasted an average of fifteen months.¹ The downturn that began in the summer of 1929 had lasted for fifteen months. A rapid and robust recovery was anticipated. In November 1930, however, a series of crises among commercial banks turned what had been a typical recession into the beginning of the [Great Depression](#).
- When the crises began, over 8,000 commercial banks belonged to the Federal Reserve System, but nearly 16,000 did not. Those nonmember banks operated in an environment similar to that which existed before the Federal Reserve was established in 1914. That environment harbored the causes of banking crises.
- [Reference: Federalreservehistory.org](https://www.federalreservehistory.org)



Federal Reserve Bulletin, Sept. 1937 Table 1, p. 868.

The Federal Funds Rate

- On any given day, banks with insufficient reserves can borrow from banks with excess reserves.
- The interest rate on these loans is the **federal funds rate**.
- The FOMC uses OMOs to target the fed funds rate.
- Changes in the fed funds rate cause changes in other rates and have a big impact on the economy.

The Federal Funds Rate

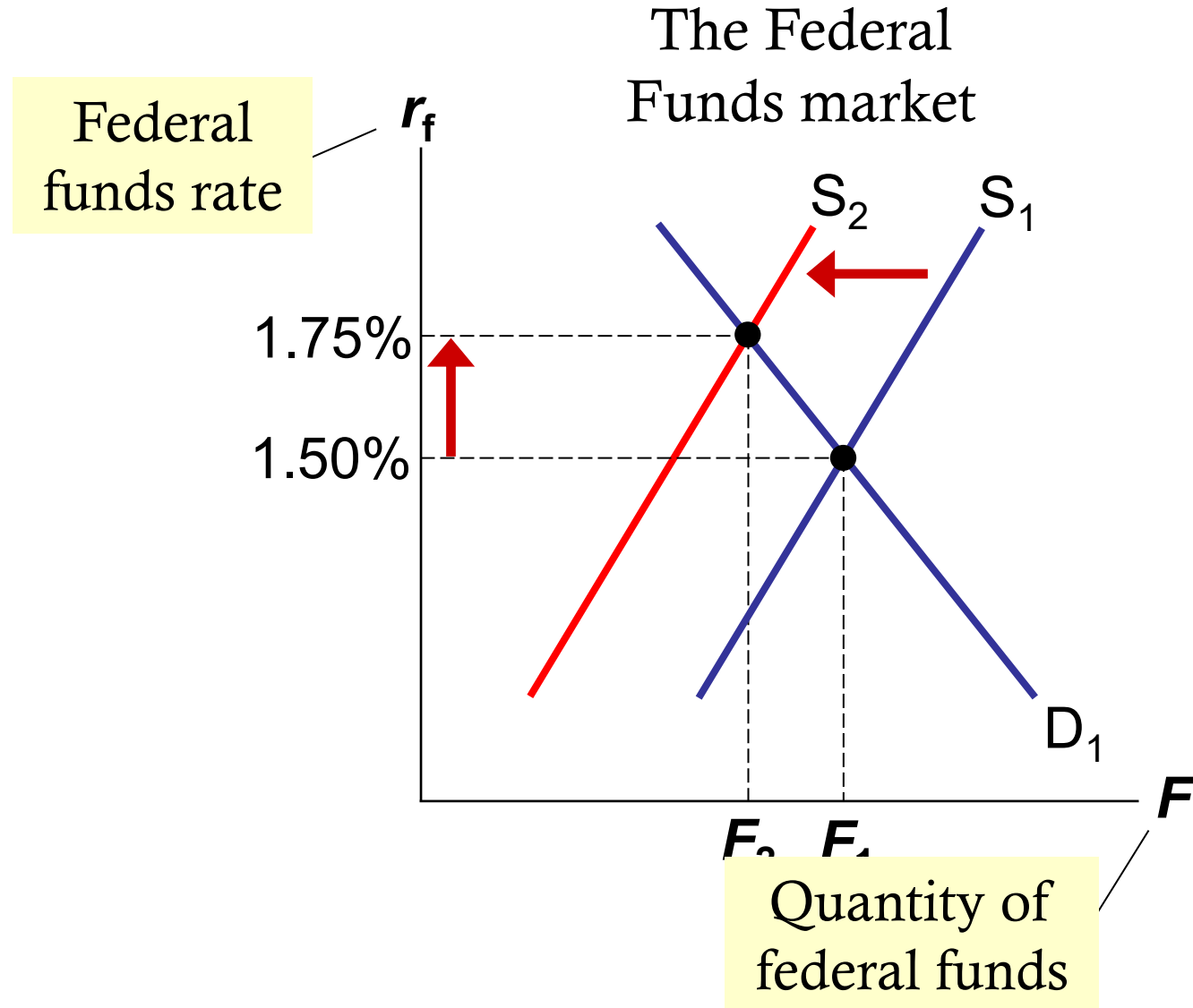


Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

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Monetary Policy and the Fed Funds Rate

- To raise fed funds rate, Fed sells government bonds (OMO).
- This removes reserves from the banking system, reduces supply of federal funds,
- causes r_f to rise.



Reflection

- Why might the Fed intentionally raise interest rates even if it slows growth?
- Consider inflation, employment, and long-run stability.

Key Takeaways

- Money has value because people trust it and it is limited.
- The Fed oversees the monetary system and controls the money supply.
- Banks create money through lending.
- The Fed uses policy tools to stabilize prices and employment.
- Its control is strong, but behavior and expectations matter.

Knowledge Check

1. If banks can create money by lending, why can't they simply make everyone wealthier?
2. A bank receives a new deposit of \$5,000 and the required reserve ratio is 10 %. What is the maximum amount of new money that can be created in the banking system?
3. Explain how open-market operations affect the money supply when the Federal Reserve buys government securities.
4. A bank has \$100,000 in checkable deposits and is required to keep 20 % as reserves. If the bank currently holds \$30,000 in reserves, how much can it lend out?

Knowledge Check Answers

1. Banks create money in the form of deposits, not real goods or services. Creating more deposits increases the money supply, but not actual wealth—it simply changes the composition of financial assets.
2. Money Multiplier = $1 / r = 1 / 0.10 = 10$
Potential increase = $\$5,000 \times 10 = \$50,000$ in total money (including the original deposit).
3. When the Fed buys bonds, it credits banks' reserve accounts, increasing reserves. Banks then make more loans, expanding deposits and the overall money supply.
4. Required reserves = $0.20 \times \$100,000 = \$20,000$
Excess reserves = $\$30,000 - \$20,000 = \$10,000$
The bank can lend \$10,000.